

$$\underline{\text{batch}} \rightarrow \{(I_1, \tau_1), (I_2, \tau_2), \dots, (I_N, \tau_N)\}$$

S_{ij} = similarity in i th image and j th caption.

$$\Rightarrow S_{ij} = \cos(I_i, \tau_j)$$

$$P(\tau_j | I_i) = \frac{\exp(S_{ij} / \tau)}{\sum_{k=1}^N \exp(S_{ik} / \tau)}$$

$$\underline{P(i|i)} = \frac{e^{S_{ii}/\tau}}{\sum_{k=1}^N e^{S_{ik}/\tau}} \approx \underline{1} \text{ (should be)}$$

probability that caption i is selected for the i th image.

in ML, we always try to minimize loss instead of increasing probability.

$$\begin{aligned} L_i &= -\log P(i|i) \\ &= -\log \left(\frac{e^{S_{ii}/\tau}}{\sum_{k=1}^N e^{S_{ik}/\tau}} \right) \end{aligned}$$

$$\sqrt{\sum_{k=1}^N e^{S_{ik}/\tau}}$$

InfoNCE loss for i th image.

$$\Rightarrow L_i = - \left[\log e^{S_{ii}/\tau} - \log \left(\sum e^{S_{ik}/\tau} \right) \right]$$

$$\Rightarrow L_i = -\frac{S_{ii}}{\tau} + \log \left(\sum_{k=1}^N e^{S_{ik}/\tau} \right)$$

if $S_{ii} \gg S_{ik}$, then the summation is dominated by S_{ii} .

$$\Rightarrow -\frac{S_{ii}}{\tau} + \log(\dots) \approx 0$$

minimized by τ

temperature (τ)

this is a scaling factor that controls the confidence of softmax.

$\tau \rightarrow 0$ means largest similarity dominates softmax.

because ^{large} exponentials $\rightarrow \infty$ and others $\rightarrow 0$.

Here, if a wrong one is large, then the wrong captions dominates.

thus, the correct similarity **MUST BE** the largest, no matter by what extent.

$\tau \rightarrow \infty$ becomes constant

$$\tau \rightarrow \infty \Rightarrow \text{every exponent} \downarrow e^{s_{ij}/\tau} = e^0 = 1.$$

$$\Rightarrow L_i = -0 + \log N = \log N$$

here all captions get same probability, no learning at all.

Gradients

$$L_{in} = -S_{ii} \quad \text{and} \quad \frac{\partial L_{in}}{\partial s_{ij}/\tau}$$

$$-\frac{1}{\tau} + \log(\sum e^{-\dots})$$

$$\frac{\partial L_{i2t}}{\partial s_{ij}} = \frac{P(j|i) - 1}{\tau}$$

Bad model

$$P_{ii} = 0.2$$

$$\text{gradient} = -\frac{0.2}{\tau}$$

$$s_{ii} \uparrow \Rightarrow P(i|i) \uparrow$$

good model

$$P_{ii} \approx 0.99$$

$$\text{gradient} = -0.01/\tau$$

not a lot of change.

$$\star \frac{\partial L_{i2t}}{\partial s_{ij}} = \frac{P(j|i) - 1}{\tau} \quad \therefore \frac{\partial}{\partial s_{ij}} \left(-\frac{s_{ii}}{\tau} \right) \rightarrow 0$$

example

$$P(j|i) = 0.3$$

$$\Rightarrow \frac{\partial L}{\partial s} = \frac{0.3}{\tau} > 0$$

$$\Rightarrow \underline{s_{ij} \downarrow}$$

which is what we want.